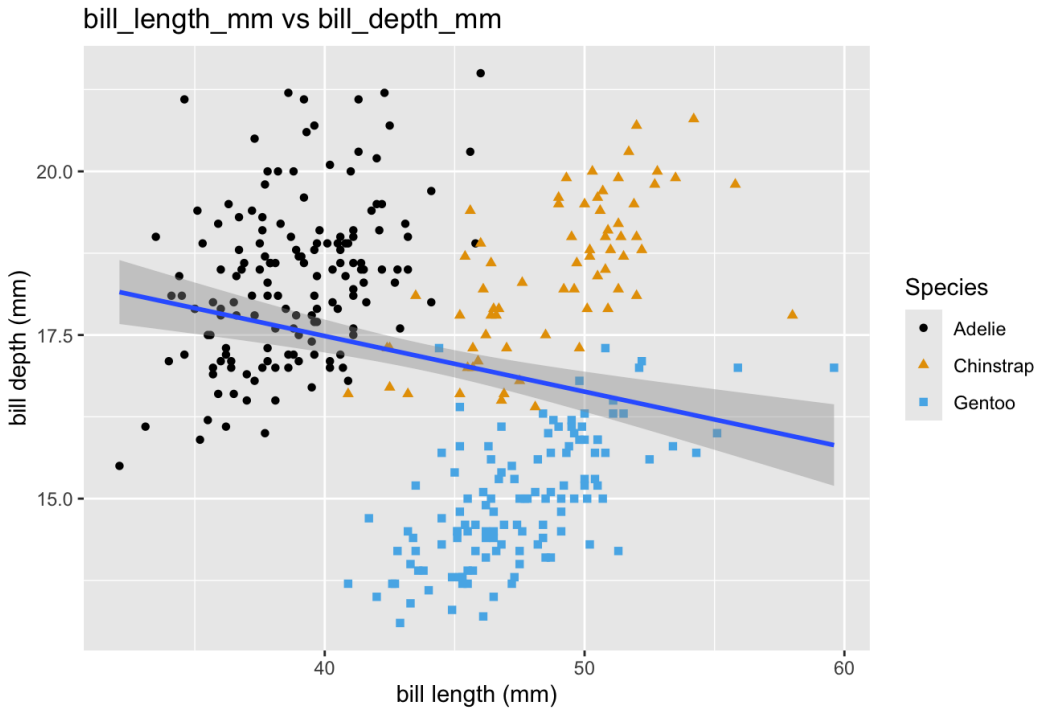
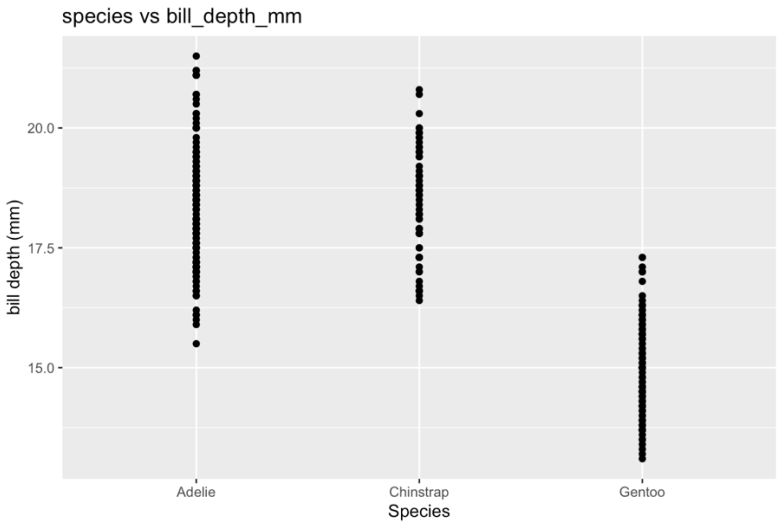
Exercise Unit 1

1- In the penguins dataset, there are 344 rows and 8 columns.
2- The bill_depth_mm variable represents bill depth measured in millimeters.
3-



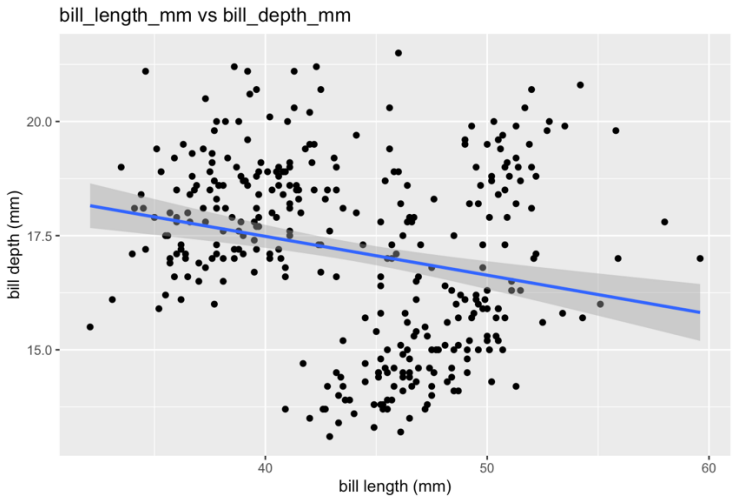
The plot shows a negative relationship between the two variables (bill length and bill depth). As bill length increases, bill depth tends to decrease. The relationship appears moderate, as there is a noticeable degree of scatter. Gentoo penguins tend to have longer, shallower bills; Adelie penguins have shorter, deeper bills; and Chinstrap penguins fall in between, with relatively long bills.

4- If we create a scatter plot of species vs bill depth, the result is the following:



The points in the scatterplot are all very close together and overlapping. In this case, a good choice for visualisation could be a box plot.

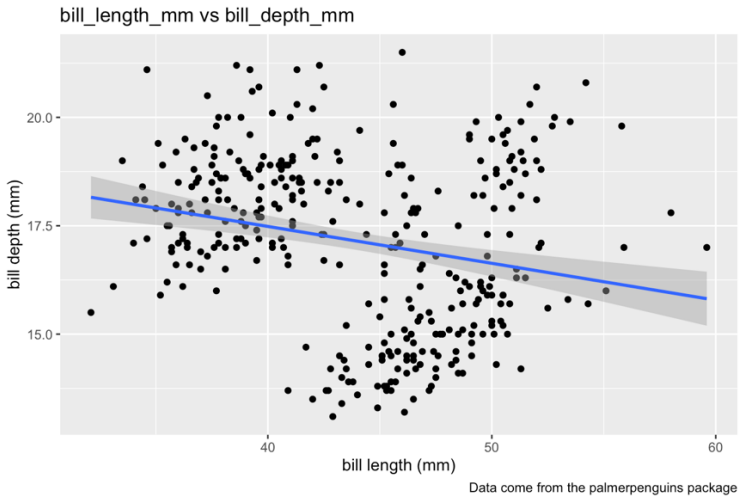
5- The na.rm argument controls how NA values are handled. By default, na.rm = FALSE, meaning that ggplot2 removes NAs and excludes observations with NA values from the analysis, but generates a warning. If na.rm = TRUE, the warning is suppressed. Scatter plot with na.rm= TRUE



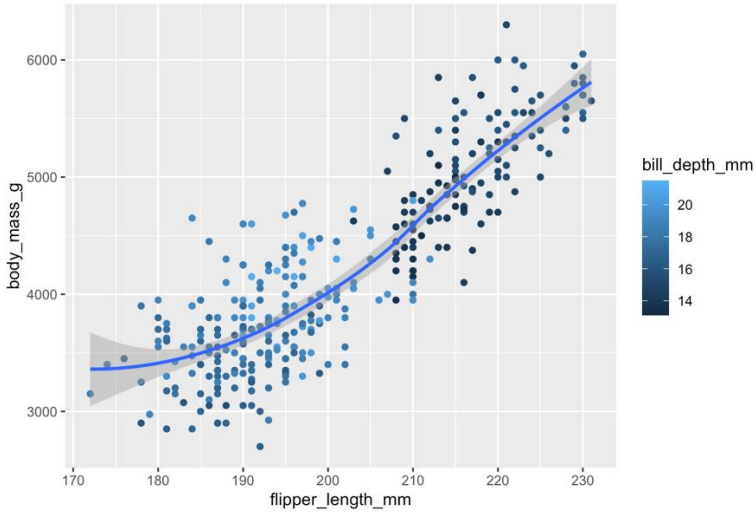
6- The code ggplot(data= penguins) + geom_point() produces an error because the x and y variables are not defined inside aes().

7- The na.rm argument specifies how geom_point() handles NA values in the x and y variables. If na.rm = FALSE, R will remove NA values but produce a warning; if na.rm = TRUE, NA values are removed without giving a warning.

8-



9-



It should be mapped at the geom level so that the line remains separate and is computed using all the data.

10-The scatter plots are identical; the difference lies in the code. In the first case, the information is specified once in the ggplot () call, whereas in the second case, it is left empty, and the information is defined within each layer.

Exercise Unit 2

1- The main differences are two. First, the plot is rotated, which is particularly useful when the categories have long names, as it provides enough space for them to be read clearly. Second, when bars are horizontal, the focus shifts more towards the ranking of categories rather than the height of the bars.

2- The correct code is:

```
library(tidyverse)
ggplot(data=mpg) +
geom_point(mapping = aes( x = displ, y=hwy)) +
```

geom_smooth(method = “lm”)

3- If I press Option + Shift + K, it produces an R Markdown file (Knit) that takes the .Rmd file, which contains both text and R code, and executes all the code to generate a final document. The same result can be achieved in RStudio by clicking on File and then Knit Document.
4- It will save the bar plot because ggsave() specifies plot = my_bar_plot. If the plot is not specified, it will save the most recently generated graph.